

Discrimination through Experience; A Contextual Bandit Problem

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Abstract

This paper aims to construct an Agent Based Model of the labour market via a contextual bandit problem. The paper seeks to uncover mechanisms by which discrimination in the labour market occurs, and subsequently, policy that can remedy said mechanisms. While the model does not fully explain discrimination in the labour market, reflections upon its results provide useful insight.

The contribution of this paper is to show that by not creating a model of equilibrium, wage distributions can be generated the more realistically replicate empirical reality, and thus provide further insight on the consequences of Experience Based Discrimination

Introduction

Discrimination in the labour market is multi-faceted, and presents itself in many forms. Discrimination can be bare-faced in its prejudice; it seeps in to the orifices of society, creating differences between groups. Yet these aren't the types of discrimination usually referred to in economics. Within academic economics, the form of discrimination that matters is the one where individuals with the same productivity are paid differing wages.

Literature Review

Discrimination

Theory

Lepage (2024) develops a model of Experience Based Discrimination. The paper, which largely motivates the development of the model presented in this

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paper, posits that firms are not only uncertain over individual productivities, but also the productivity of the group. Typically economic literature has compartmentalised discrimination into two main categories, *taste-based*, and *statistical* discrimination. The former, associated with Becker (1957), corresponds to discrimination purely because the hiring manager prefers (or dislikes) some group. The latter is predicated on the assumption that firms are not inherently discriminatory, but are discriminatory in practice due to failures in ability to read signals. The Statistical Discrimination literature is associated with the work of Arrow (1973) and Phelps (1972). Here, discrimination arrives because of "*scarcity of information about the existence and characteristics of workers and jobs*". Thus, the 'Experience Based Discrimination' is closely related to the statistical discrimination literature. However, there is a large difference, as noted by Lepage (2024), Statistical Discrimination relies on some inherent difference in the characteristics of groups, while Experience Based Discrimination does not.

Other models such as that of Lundberg and Startz (1983) utilise endogenous human capital decisions to engender discrimination; if workers from minority groups expect to receive a lower wage for their efforts, then they will underinvest in human capital, creating a divergence in productivity between groups. The model, however, relies on innate differences on reliability of observable test scores upon which wages are based. and Coate and Loury (1993), show that perverse employer beliefs can cause minority groups to underinvest in human capital, which creates divergence between group productivities.

Empirics

Li et al. (2020) use firm-level recruitment data to test ML algorithms in hiring policy. Due to a lack of data on performance, the data objective function is to maximise hiring potential conditional on being interviewed; they find that algorithms that understand the trade-off between exploration and exploitation perform much better on metrics of fairness and efficiency of hiring policy than 'supervised' ML algorithms focus on exploitation. Lepage (2024) uses an experimental labour market design to observe experience based discrimination; they too find evidence that endogenous learning can create discrimination. Both of these papers set up their empirics as a 'Multi-Armed Bandit' problem.

Multi Armed Bandit Problems

Multi Armed Bandit (MAB) problems have a strong history in economics; MABs are usually applied to problems characterised by a trade-off between exploration and exploitation Bergemann and Välimäki (2006), that is, executing an action we currently believe to be optimal, versus exploring the action space to understand whether some other action is optimal. The Contextual bandit problem is similar, but the agent observes a an N-dimensional *feature* vector, providing context over each arm.

Reinforcement Learning

Agent Based Modelling

As noted by Lundberg and Startz (1983), models of competitive markets and perfect information are unable to explain how discrimination can arise. This is a problem common to many economic models in some capacity, and can only provide treatment of such firm behaviour as an exogenous shock, this has been a great motivation for the literature on Agent Based Models (ABMs). Pangallo and del Rio-Chanona (2024) provides a definition of ABMs.XXX and XXX provide examples of ABMs in labour markets. As noted by XXX, ABMs receive plaudits for their ability to explain endogenous behaviour, however mainstream economists tend to avoid them due to the perceived arbitrariness of modelling decisions.

Model

Workers

There are N_w workers in the set of workers W , each belonging to a group $G \in g = 1, 2, \dots, G$ with proportion N_g/N_w in the population, such that $\bigcup_{g=1}^G G_g = W$, and $G_i \cap G_j = \emptyset \quad \forall i, j \in \{1, 2, \dots, G\}$ where $i \neq j$.

Workers have an innate ability, which constitutes their marginal productivity, but is unobservable to firms, and a purely stochastic signal that is observable to the firms.

$$\text{Productivity: } x_i \sim \mathcal{N}(\mu_g, \sigma_{pg}^2)$$

$$\text{Signal: } s_i \sim \mathcal{N}(0, \sigma_{sg}^2)$$

This signal could represent the qualifications of a worker, or the results of some test set by the employer. Either way, the fact it has non zero variance represents the reality that for some exogenous reasons, the signals observed by employers do not perfectly represent the individual's ability, and therefore productivity. With a mean of 0, the signal on the workers' productivity is unbiased.

Then:

$$x_i = \mu_g + \epsilon_{pg}$$

$$s_i = x_i + \epsilon_{sg} = \mu_g + \epsilon_{pg} + \epsilon_{sg}$$

Each group of workers are assumed to have homogenous distributions of productivities and signals. Unlike other models of statistical discrimination, the nature of endogenous learning within the model does not necessitate the groups of workers to be inherently different; *workers* are heterogeneous, *groups* are homogenous.

Then:

$$\mu_g = \mu, \quad \sigma_{pg}^2 = \sigma_p, \quad \sigma_{sg}^2 = \sigma_s \quad \forall g$$

This is motivated by the idea of economic discrimination, which has principally been concerned with discrimination between groups which all inherently contain heterogenous individuals, but could be equal on a meta level. The original measurement of discrimination employed by ‘?’ sought to observe the lowest productivity individual employed. Here, the data will provide insight in ot wage distributions amongst groups, employment levels, and wage offers.

Firms

There are N_f firms in the market, each facing a ‘Multi Armed Bandit’ problem; at each time period t , the firm matches with a number of prospective employees from the worker group, and they must decide which ‘arm’ (worker) to employ for the coming period. The firm must pull the arm (employ the worker) that they believe will provide them with the greatest profit. The problem becomes a ‘Contextual Bandit’ problem because the firm observes a *feature vector* describing the nature of the prospective workers: they observe both the group they belong to, and the signal on their productivity.

The firms are modelled as *Bayesian Updaters*. The firms model both the distribution of the productivities and signals, and the distributions of the parameters that govern these distributions.

Productivity:

$$p(x_i \mid \mu_g, \sigma_{pg}) \sim \mathcal{N}(\mu_g, \sigma_{pg})$$

$$\pi(\mu_g, \sigma_{pg} \mid x_i) \sim \mathcal{NIG}(\mu_{g0}, \eta_{g0}, \alpha_{g0}, \beta_{g0})$$

Signal:

$$p(s_i \mid \sigma_{sg}) \sim \mathcal{N}(0, \sigma_{sg})$$

$$\pi(\sigma_{sg} \mid x_i) \sim \mathcal{IG}(\delta_{g0}, \kappa_{g0})$$

For productivity, the firms have uncertainty over both the mean and the variance of the true distribution. This motivates the use of the *Normal Inverse Gamma* distribution, which is a conjugate prior to the normal distribution where there is uncertainty over both the mean and variance parameters. A prior is a *conjugate prior* for a likelihood function if they are in the same parameterised family, that is: $\{\pi(\mu_g, \sigma_{pg} \mid x_i), p(x_i \mid \mu_g, \sigma_{pg})\} \in \mathcal{F}$, where $\mathcal{F}$ is a closed set (Murphy, 2022). For the signal they only have uncertainty over the variance, and correctly assume that the signal is unbiased with mean 0; for analogous reasons, the *Inverse Gamma* is used a prior for the distribution of the variance parameter [See Appendix A for a full explanation and derivation].

To obtain a ‘predictive distribution’ for the productivity of the next individual from each group, one can integrate out over the joint posterior. This gives a student’s- t distribution^{*†}(Ramírez-Hassan, 2023):

$$x^* \mid \mathbf{x} \sim t_{2\alpha_{gn}} \left(\mu_{gn}, \frac{\beta_{gn}(\eta_{gn} + 1)}{\alpha_{gn}\eta_{gn}} \right) \quad (1)$$

Although the groups have identical parameters, firms have unique hyperparameters (the parameters that determine the shape of the prior distributions) for each worker group; Figure 1 shows examples of how these hyperparameters lead to differing prior beliefs.

When a firm hires a worker, they observe their productivity, they subsequently update their posterior belief on the productivities and signals from that group:

$$\begin{aligned}\kappa_{g(n+1)} &= \kappa_{gn} + 1 \\ \mu_{g(n+1)} &= \frac{\kappa_{gn}\mu_{gn} + x}{\kappa_{gn} + 1} \\ \alpha_{g(n+1)} &= \alpha_{gn} + \frac{1}{2} \\ \beta_{g(n+1)} &= \beta_{gn} + \frac{\kappa_{gn}(x - \mu_0)^2}{2(\kappa_{g(n+1)})}\end{aligned}$$

We can decompose the updates for μ_{g1} in to two terms:

$$\mu_{g(n+1)} = \mu_{gn} \left(\frac{\eta_{gn}}{\eta_{gn} + 1} \right) + x_i \left(\frac{1}{\eta_{gn} + 1} \right)$$

Which is a weighted average of the prior on the mean of productivity, and the new data observed on the worker i from group g . This provides insight in to how the firms belief system works and develops; as the firm observes more data, the prior (posterior) distributions of the hyperparameters become more heavily weighted towards the data observed. Additionally, if the firm have strong priors (that is, η_{g0} large), then they will require more data to overcome the prior beliefs about the distribution of the hyperparameters.

In the model, firms are assumed to have a strong degree of monopsony power; since matches only last one period, workers cannot reject in order to continue search, as they will remain unemployed for that period if they reject. Then firms determine the wage entirely. Firms use a weight average of the expected productivity of an individual, and the observable signal. The weight is determined by the variance of each of these, as given by the firms current posterior distribution. Since the firms update their priors over each group, the wage is determined partly by their group membership: if the firm is uncertain about the group's productivity, they will more heavily weight the signal, and vice versa.

$$\begin{aligned}w_{ig} &= \phi * \mathbb{E}[\mu_g] + (1 - \phi) * s_i \\ \phi &= \frac{\mathbb{E}[\sigma_{sg}]}{\mathbb{E}[\sigma_{sg}] + \mathbb{E}[\sigma_{pg}]}\end{aligned}$$

* Where μ_{gn} is the posterior belief of μ for group g in period n .

† The degrees of freedom $2\alpha_{gn}$ increase with the number of observations n , so the predictive distribution converges to a Normal as $n \rightarrow \infty$.

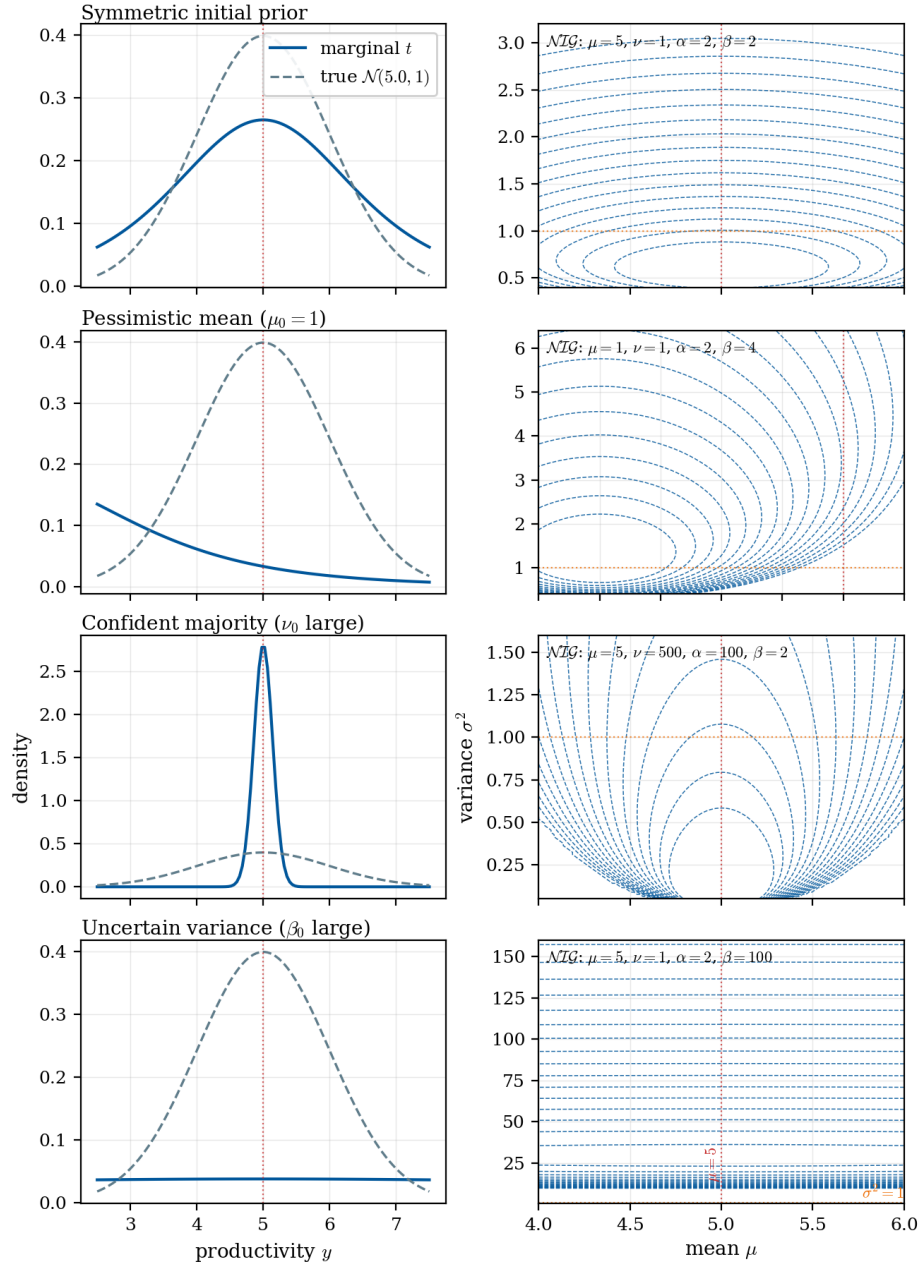


Figure 1: Examples of prior beliefs for varying hyperparameters. The left hand side shows the student's- t (predictive) distribution [see Equation 1], while the right hand side shows the corresponding Normal Inverse Gamma joint posterior over the mean and variance.

Then as $\sigma_{pg} \rightarrow \infty, \phi \rightarrow 0$ and we put no weight on the group's productivity distribution. While as $\sigma_{pg} \rightarrow 0, \phi \rightarrow 1$ and we put full weight on the group's productivity because firm believes the productivity of the group has a mass fully centered on the mean.

And because these variances are modelled by Inverse Gamma distributions, we use the expected values of these distributions to obtain the variances, where parameters are updated as in (?).

$$\mathbb{E}[\sigma_{sg}] = \frac{\beta_{gn}}{\alpha_{gn}}$$

$$\mathbb{E}[\sigma_{pg}] = \frac{\delta_{gn}}{\kappa_{gn}}$$

Utilising the observed information, and posterior distributions over each group, the firms selects worker i to maximise expected profit π :

$$\operatorname{argmax}_i \pi = \mathbb{E}(\mu_i) - w_i$$

Rejection Since workers have no reservation wage (they cannot, matches last one period and thus rejecting offers in pursuit of a better match is nonsensical), all rejections must be because another firm offered a better wage to the worker. Then, even in rejection, the firm learns about the productivity of a worker. If Firm A's offer is rejected by the worker, then they know that at least one other firm (Firm B), has offered a better wage. Then

$$\phi^A \cdot \mu_{gn}^A + (1 - \phi^A) \cdot s_i < \phi^B \cdot \mu_{gn}^B + (1 - \phi^B) \cdot s_i$$

$$\mu_{gn}^B > \frac{\phi^A \cdot \mu_{gn}^A + [\phi^A - \phi^B] \cdot s_i}{\phi^B}$$

By letting $\phi^A = \phi^B$ it is implied that $\mu_{gn}^B > \mu_{gn}^A$. Then firms can infer that other firms have had better experiences with individuals from this group; firms update their posterior belief on the group of the individual even when their offer is rejected. While firms do not directly observe the worker's productivity, they can assume that it is drawn from a truncated distribution above μ_{gn}^A , and update the posterior based on this information. The specifics of this process is left to Appendix B.

The Market

The timeline of the model is as follows:

Workers exit the market at a rate of δ , at which point they are replaced by a worker of the same group. Meanwhile, if firms consistently make negative profits, and cumulative profits drop below some threshold τ , then they cease to exist and are replaced by a new firm entering the market.

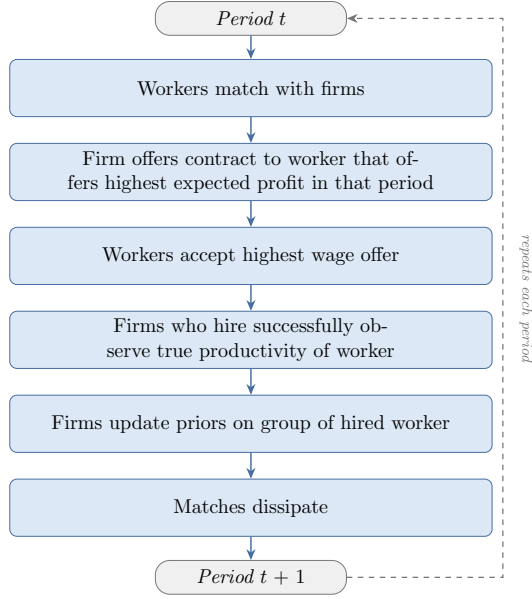


Figure 2: Timeline of the model. Each period t consists of these sequential stages which repeat $\forall t \in T$.

Simulations

Consider the case of two groups in the population $G = \{i, j\}$ with $N_j/N_w = 0.1$, and $N_w = 15,000$ workers, and $N_f = 10,000$ Firms. Then group j are the minority in the population of workers. Each firm begins with the same initial priors, which are distinct for each group.

For group j , the minority group, the firm will have uninformative priors, while for group i , they will begin with priors that are informative about the true distribution of productivity. That is, the firm are much more confident in their knowledge of the distribution of one group than another. This difference will drive much of the results in the model. Additionally, firms have equally uninformative priors on the signal for all groups. This difference in uncertainty over the productivity of each group are the only differences that create a divergence in wages between groups.

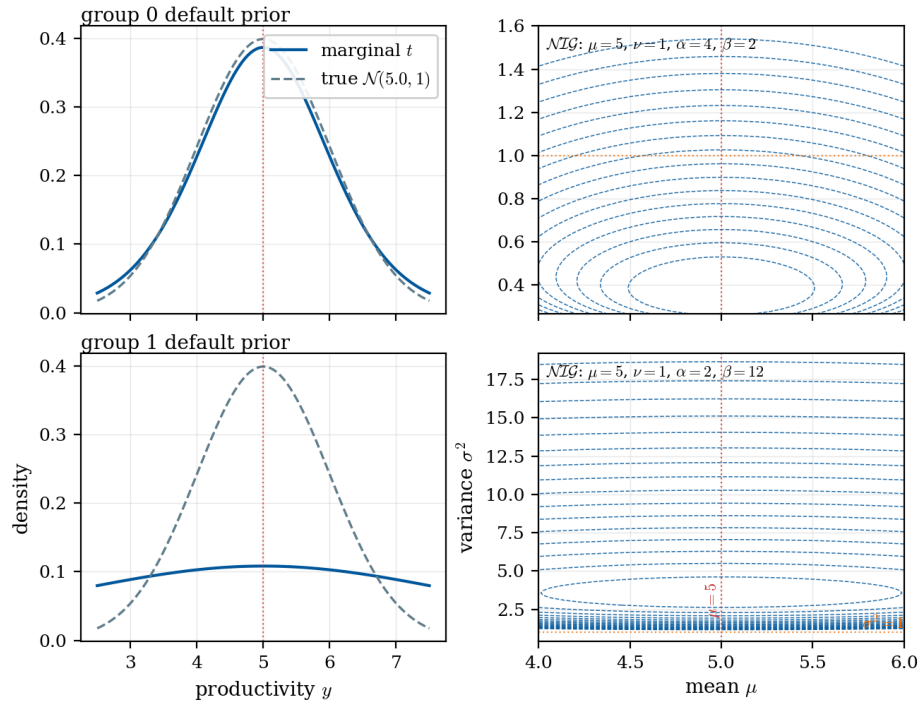


Figure 3: Initial priors for each firm for groups i, j. $\{\mu_{i0} = 5, \eta_{i0} = 1, \alpha_{i0} = 4, \beta_{i0} = 2\}; \{\mu_{j0} = 5, \eta_{j0} = 1, \alpha_{j0} = 2, \beta_{j0} = 4\}$

The Unfettered Market

Policy Intervention

Discussion

It should be of course acknowledged, that while each model purports some specific mechanism of discrimination, the true mechanism is likely to be some combination of each.

Li et al. (2020): in their paper, the contextual bandit problem utilises many aspects of heterogeneity; it could be seen that in the model presented here, how the UCB algorithm would be able to cover a multitude of aspects to increase fairness such as geography, race, gender, education: hiring such individuals would allow the firm to discern ability from background. (something about causal inference in contextual bandits)

Relating to the point on endogeneity of human capital decisions etc: if this were to be implemented in real life, then the algorithm may learn that minorities are lower productivity due to systemic bias. Li et al. (2020) do not find this to be true. Additionally one could also criticise the UCB algorithm in this case for over exploration: we know that true economic discrimination is hiring

Algorithmic Fairness

A key point that this model provides guidance on is that of Algorithmic Fairness. If employers are to use Machine Learning to decide who to hire in their hiring policy, this could lead to a bias analogous to that presented in this paper; underlying data inherently contains the kernel of human bias, and carries this forward. This highlights the importance of policy intervention, and underpins the common problem that Reinforcement Learning seeks to solve: one of *exploration vs exploitation*.

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A Bayesian Updating derivation

$$P_g | \mu_g, \sigma_{pg}^2 \sim \mathcal{N}(\mu, \sigma_{pg})$$

$$\mu | \sigma_{pg}^2 \sim \mathcal{N}(\mu_0, \frac{\sigma_{pg}^2}{\eta_0})$$

$$\sigma_{pg}^2 \sim IG(\alpha_0, \beta_0)$$

$$\pi(P_{new} | P, \mu_i, \sigma_{pg}, \alpha_0, \beta_0) = p(P)$$

The posterior distribution of μ, σ_{pg}^2 is as follows:

$$\pi(\mu_g, \sigma_{pg} | x) = \frac{p(x | \mu_g, \sigma_{pg}) p(\mu_g, \sigma_{pg})}{p(x)}$$

$$\propto p(x | \mu_g, \sigma_{pg}) p(\mu_g, \sigma_{pg})$$

$$\propto p(x | \mu_g, \sigma_{pg}) p(\mu_g | \sigma_{pg}) p(\sigma_{pg})$$

Where the prior on the joint probability distribution of the mean and variance of productivity must be decomposed in to a product of conditional and marginal distributions.

$$\begin{aligned}
p(\mu, \sigma^2 \mid x) &\propto \underbrace{(2\pi\sigma_{pg})^{-1/2} \cdot \exp((2\sigma_{pg}^2)^{-1}(x_i - \mu_g)^2)}_{\text{Likelihood of data given } \mu, \sigma_{pg}} \\
&\quad \cdot \underbrace{(2\pi\frac{\sigma_{pg}}{\eta_0})^{-1/2} \cdot \exp((2\frac{\sigma_{pg}}{\eta_0})^{-1}(\mu_g - \mu_0)^2)}_{\text{prior on } \mu_g, \text{ conditional on } \sigma_{pg}} \\
&\quad \cdot \underbrace{\frac{\beta_0^{\alpha_0}}{\Gamma(\alpha_0)} \cdot (\sigma_{pg})^{-(\alpha_0+1)} \cdot \exp(-\frac{\beta_0}{\sigma_{pg}})}_{\text{prior on } \sigma_{pg}}
\end{aligned}$$

Then we can remove any normalising terms which do not depend on the unknown parameters, and are thus normalising constants:

$$\begin{aligned}
p(\mu, \sigma^2 \mid x) &\propto \left(\frac{1}{\sigma^2}\right)^{\alpha_0+2} \exp\left(-\frac{1}{2\sigma^2} [(x - \mu)^2 + \kappa_0(\mu - \mu_0)^2 + 2\beta_0]\right) \\
&\propto \left(\frac{1}{\sigma_{pg}^2}\right)^{\frac{1}{2}} \exp(-\frac{1}{2\sigma^2}(x - \mu_{g1})^2) \left(\frac{1}{\sigma_{pg}^2}\right)^{\alpha_1} \exp(-\frac{1}{2\sigma^2}(\mu_g - \mu_{g1})^2)
\end{aligned}$$

Where:

$$\kappa_1 = \kappa_0 + 1 \quad (2)$$

$$\mu_1 = \frac{\kappa_0\mu_0 + x}{\kappa_0 + 1} \quad (3)$$

$$\alpha_1 = \alpha_0 + \frac{1}{2} \quad (4)$$

$$\beta_1 = \beta_0 + \frac{\kappa_0(x - \mu_0)^2}{2(\kappa_0 + 1)} \quad (5)$$

We can decompose the updates for μ_{g1} in to two terms:

$$\mu_{g1} = \mu_{g0} \left(\frac{\eta_{g0}}{\eta_{g0} + 1} \right) + x_i \left(\frac{1}{\eta_{g0} + 1} \right)$$

The firm must also model their beliefs about the nature of the signal for each group. It is assumed that the firm correctly believe that the signal is unbiased, as such, updates on the signal only constitute one unknown parameter (the variance).

$$\begin{aligned}
x_i | \mu_g, \sigma_{pg}^2 &\sim \mathcal{N}(\mu, \sigma_{pg}) \\
\sigma_{pg}^2 &\sim IG(\delta_{g0}, \kappa_{g0}) \\
\pi(\sigma_{pg} | x, \mu_{gs} = 0) &= \frac{p(x | \mu_{gs} = 0, \sigma_{sg})p(\sigma_{sg})}{p(x)} \\
&\propto p(x | \mu_{gs} = 0, \sigma_{sg})p(\sigma_{sg})
\end{aligned}$$

$$\begin{aligned}
p(\mu, \sigma^2 | x) &\propto \underbrace{(2\pi\sigma_{sg})^{-1/2} \cdot \exp((2\sigma_{sg}^2)^{-1}x_i^2)}_{\text{Likelihood of data given } \sigma_{sg}} \\
&\quad \cdot \underbrace{\frac{\kappa_0^{\delta_0}}{\Gamma(\delta_0)} \cdot (\sigma_{pg})^{-(\delta_0+1)} \cdot \exp(-\frac{\kappa_0}{\sigma_{sg}})}_{\text{prior on } \sigma_{sg}}
\end{aligned}$$

$$p(\sigma^2 | x) \propto \left(\frac{1}{\sigma^2}\right)^{1/2} \exp\left(-\frac{x^2}{2\sigma^2}\right) \cdot \left(\frac{1}{\sigma^2}\right)^{\alpha_0+1} \exp\left(-\frac{\beta_0}{\sigma^2}\right) \quad (6)$$

$$\propto \left(\frac{1}{\sigma^2}\right)^{\alpha_0+3/2} \exp\left(-\frac{1}{\sigma^2} \left[\beta_0 + \frac{x^2}{2}\right]\right) \quad (7)$$

With updates:

$$\alpha_1 = \alpha_0 + \frac{1}{2}$$

$$\beta_1 = \beta_0 + \frac{x_i^2}{2}$$

B Updating Posterior Beliefs Upon Rejection

C Robustness checks